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SHORT COMMUNICATION

Extraction of rare earth elements from NdFeB scrap by AlF_3–NaF melts

Z. S. Hua*[1], L. Wang[1], J. Wang[1], Y. P. Xiao[1,2], Y. X. Yang[1,3], Z. Zhao[1] and M. J. Liu[1]

To conserve both energy and rare earths, a new technology for recycling of NdFeB magnet scrap was proposed in this paper based on the selective fluorination with AlF_3–NaF melts. After fluorination, the reaction products were examined by scanning electron microscopy equipped with energy dispersive X-ray spectrometry, X-ray diffraction and inductively coupled plasma. Rare earth elements in the scrap were selectively converted to rare earth fluorides in the yield of ca. 94·2%, leaving Fe, B and impurities in the solid residue. The differences in chemical activity between Fe, impurities and rare earth should be mainly responsible for the selective fluorination process. The obtained fluorides can be subsequently processed with molten salt electrolysis for the direct production of Al–Nd based alloy.

Keywords: NdFeB scrap, Rare earth, Fluorides, Extraction, Recycling

Introduction

NdFeB permanent magnet has been widely used in various fields owing to its excellent magnetic properties since it was developed in 1983.[1] Currently, the average annual growth in NdFeB magnet market has increased to 70% due to the increase in world demand.[2] During the magnet manufacturing process, approximately 30 wt-% scrap is generated, because of the cutting, grinding and polishing operations. Additionally, the amount of end-of-life scrap is also increasing year by year. It is well known that rare earth elements (REEs) play an important role in modern industrial activities and their insignificance in high technology continues to grow.[3] Therefore, to establish an efficient extraction and recycling technology for the recovery of REEs from NdFeB scrap becomes the hot issue from the point of view of environmental protection as well as resources conservation.

Presently, NdFeB scrap is generally recovered via hydrometallurgical technology. Several wet processes including acid dissolution,[4] ion exchange or solvent extraction[5,6] and oxalate precipitation[7] are used for REEs recovery. However, theses processes involve the generation of much waste acid and effluent. Furthermore, to meet the requirements of industrial application, the recovery process should have as few steps as possible, thus facilitates to lower the processing cost. Several other dry processes[8–15] have also been proposed for recycling of NdFeB scrap, but not met the practical use yet in current stage. A recycling technology using molten LiF–NdF_3 has been reported.[16] However, this method is just suitable for removing rare earth oxides from NdFeB waste.

Based on the background, a novel recycling method using AlF_3–NaF melts as the extractant was adopted to recover REEs from NdFeB scrap. Owing to the selective fluorination, both of REEs and their oxides in the magnet scrap can react with AlF_3, forming rare earth fluorides, while Fe, B and the impurities (Co, Cu, Zr, etc.) are not fluorinated and essentially immiscible in AlF_3–NaF melts. Consequently, REEs are readily separated as rare earth fluorides from Fe based metallic solid residue.

The extracted rare earth fluorides in the melts can be processed with molten salt electrolysis for the direct production of Al–Nd based alloys in LiF–CaF_2 medium on tungsten electrode,[17] and then rare earth metals can be obtained by evaporating aluminium out of the Al–Nd liquid alloys.[14] This study merely focuses on the extraction process for REEs.

Experimental

Off-specification NdFeB magnet was crushed to granules with diameter of 3–5 mm and served as the scrap sample for the experiments, and the composition is shown in Table 1. Powder AlF_3 (99% purity) and NaF (99% purity) were used for preparing fluoride melts, previously dehydrated under vacuum for more than 72 h at 300°C to remove excess water. The fluorides were mixed to composition of NaF–50mol.-%AlF_3 and charged in a graphite crucible. Then, the magnet granules were separately imbedded in the fluorides mixture. After being heated at 800–1100°C for 1–6 h in an Ar gas flow to protect dehydrated fluorides from deliquescing, the resultant fluorides and residual scrap were ground into powders with size less than 75 μm subjected to analysis.

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Table 1 Elemental compositions of NdFeB magnetic scrap

Element	Fe	Nd	Pr	Dy	B	Al	Co	Zr	Cu	Si
Content/wt-%	65·71	23·97	4·42	2·75	1·01	0·86	0·80	0·15	0·20	0·13

The reaction products were examined by scanning electron microscopy (SEM: SSX-550, Shinadzu Corporation) at an accelerating voltage of 15 kV. Chemical analysis was performed simultaneously with SEM using energy dispersive X-ray spectroscopy (EDS). The constitution phases were determined by X-ray diffraction (XRD) measurements using Cu K_α radiation, and the elemental content was analysed using inductively coupled plasma atomic emission spectrometry (ICP-AES). Rare earth recovery rate R was calculated from the following formula

$$R = 1 - \frac{w^{\mathrm{RE}}_{\mathrm{residue}}}{w^{\mathrm{RE}}_{\mathrm{initial}}} \times 100\% \quad (1)$$

where $w^{\mathrm{RE}}_{\mathrm{residue}}$ is mass of rare earths Nd, Pr and Dy in residual scrap after experiment, and $w^{\mathrm{RE}}_{\mathrm{initial}}$ is the initial mass of rare earths in NdFeB scrap. Data for recovery rate were gathered on three experiments.

Results and discussion

Extraction of REEs (Nd, Pr, Dy) from NdFeB scrap by molten fluorides are based on the following selective fluorination of AlF_3 in the fluoride melts

$$Nd + AlF_3 = NdF_3 + Al \quad (2)$$

$$Pr + AlF_3 = PrF_3 + Al \quad (3)$$

$$Dy + AlF_3 = DyF_3 + Al \quad (4)$$

Owing to the differences in chemical activity between Al and Fe, impurity elements including Cu, Al, Co, Zr, Si, etc., these impurities as well as Fe in the scarp can not be fluorinated by the fluoride melts.

By comparing the content of Nd, Pr, Dy in the initial scrap and residue, it was demonstrated that REEs in NdFeB scrap can be recycled by selective fluorination of AlF_3–NaF melts. The rare earth recovery rate under different experimental conditions was gathered in Table 2. It indicates that the rare earth recovery rate increase from 71·1 to 94·2% when the temperature was raised from 800 to 1100°C. Fluorination temperature shows a remarkable effect on rare earth recovery rate. This is probably attributed to the melts viscosity. Generally, the viscosity of molten fluorides decreases while raising temperature and thus the diffusion of reactant and product in the melt is accelerated, which is favourable to the REEs fluorination process. Moreover, the Gibbs free energy of reactions between REEs and AlF_3 declines with the temperature increasing, which implies that high temperature is beneficial for the fluorination thermodynamically.

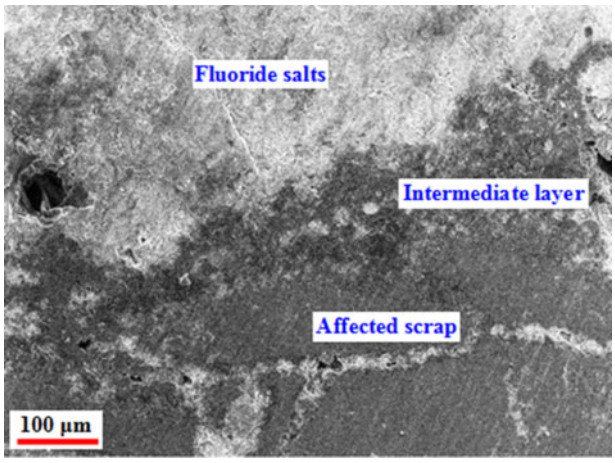
There is a considerable improvement in the recovery rate with holding time increasing from 1 to 3 h, but only a slight increase by further elongating holding time. As the reaction progressed, the concentration gradient between the reaction interface and the bulk melts was declining owing to the decrease of reactant concentration and the increase in product concentration of the melts, which resulted in the decrease in reaction rate. Furthermore, the surface area of the scrap increases during fluorination process,[8] which accelerates the fluorination to chemical equilibrium. Therefore, the reaction time of 3 h was enough to accomplish the nearly complete selective fluorination for REEs in the granule scrap.

Generally, high fluorides/scrap mass ratio is beneficial for the fluorination. Meanwhile, to achieve complete recycling of REEs, the amount of AlF_3–NaF melts must be enough to cover the magnet scrap entirely. According to REEs content in the initial magnet scrap, about 2 g AlF_3 is required for complete fluorination of 10 g NdFeB scrap. In the experiment series, the amount of AlF_3 in the mixed fluorides was ca. 20 g, which was $\sim$10 times of equivalent amount. Deducting loss from volatilisation at high temperature, the amount of AlF_3 was still excessive for complete fluorination of REEs in the scrap. Consequently, after fluorination at 900°C for 3 h, the rare earth recovery rate just showed a slight variation in the range from ca. 82–83% while the fluorides/scrap mass ratio increased.

Figure 1 shows the magnification micrographs of the resultant product after fluorination at 900°C for 2 h. Close observation of the sample reveals that the sample consists of three distinct phases: reacted scarp NdFeB, intermediate layer and fluoride salts. It is clear that molten fluorides have diffused into the scrap at reaction temperature. The compositions of these phases

Table 2 Rare earth recovery rate from NdFeB scrap by using molten fluorides

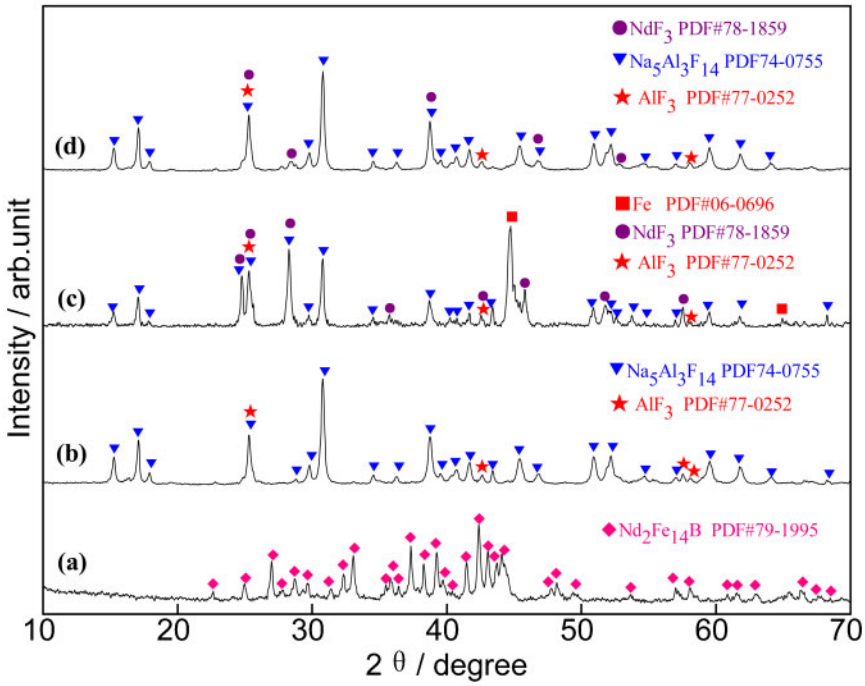
Fluorination temperature/°C	Holding time/h	Mass of scrap/g	Mass of fluorides/g	Recovery rate/%
800	3	5·008	30·011	71·1
900	3	5·006	30·009	82·9
1000	3	5·002	30·005	89·7
1100	3	5·005	30·006	94·2
900	1	5·003	30·007	74·3
900	2	5·005	30·008	77·5
900	4	5·009	30·012	83·5
900	5	5·007	30·009	84·1
900	6	5·007	30·005	84·5
900	3	6·003	30·003	82·8
900	3	8·006	30·008	82·5
900	3	10·002	30·010	82·3



1 SEM image of resultant product after fluorination at 900°C for 2 h

gressed, Nd in the reacted scrap incessantly converted to neodymium fluoride and diffused into AlF_3–NaF melts, leaving the unreacted Fe in the intermediate layer. Overall, Nd in the scarp was fluorinated to form NdF_3 and dissolved in the melts, resulting in the generation of Nd element in fluoride salts, while unreacted residue, which was mainly composed of Fe, remained in the intermediate layer, as shown in Fig. 1 and Table 3.

Figure 2 shows the XRD patterns of the initial reactants and the obtained products after fluorination at 1000°C for 6h. The XRD pattern of the initial magnet scrap was assigned as the tetragonal phase of $Nd_2Fe_{14}B$ (curve *a*). In this experiment, the mole ratio of AlF_3/NaF was ~1 : 1, and the molten fluorides mixture was crystallised to form the tetragonal phase of $Na_5Al_3F_{14}$ and hexagonal phase of AlF_3 during cooling down to room temperature. Therefore, diffraction peaks of $Na_5Al_3F_{14}$ and AlF_3 were observed in the initial fluoride melts



a initial magnet scrap; *b* initial molten fluorides; *c* residual scrap after experiment; *d* obtained salts

2 X-ray diffraction patterns of reactants and products

determined by EDS are shown in Table 3. Note that owing to the small amount in NdFeB magnet, B, Pr, Dy and the impurity elements have not been detected by EDS. By comparing the composition of the initial scrap (Table 1) and resultant product, it confirms that selective fluorination on NdFeB scrap does generate, which results in an Nd depleted region existing in the reacted scrap where diffusion of the AlF_3 from the fluoride melts has occurred. As the fluorination pro-

Table 3 EDS results of different phases in Fig. 1

	Composition/wt-%				
Phase	Fe	Nd	F	Al	Na
Reacted scrap	90·32	9·68	…	…	…
Intermediate layer	100	…	…	…	…
Salts	…	10·48	55·76	20·15	13·61

(curve *b*). After the selective fluorination, diffraction peaks of cubic Fe were detected from the residual scrap (curve *c*). Meanwhile, peaks of $Na_5Al_3F_{14}$, AlF_3 and NdF_3 were also identified in the residue, as shown in curve c, and these were derived from the molten fluorides deposited on the surface of the granule scrap. Except $Na_5Al_3F_{14}$, AlF_3 and NdF_3, other two rare earth fluorides (PrF_3 and DyF_3) supposed to appear have not been found due to their low level of content in the obtained salts (curve *d*). These results indicate that REEs in NdFeB magnet scrap have been selectively extracted out and converted to rare earth fluorides dissolving in the AlF_3–NaF melts, leaving Fe-based metallic solid in the residual scrap. These are in good agreement with the results of EDS analysis.

Owing to their chemical activity, the REEs on NdFeB scrap surface were inevitably oxidised into rare earth oxides. To clarify the fluorination mechanism on the magnet scrap, the individual fluorination experiment of

pure Nd and Nd_2O_3 was carried out at 900°C for 3 h. The fluorination proceeded thoroughly on both Nd and Nd_2O_3 to generate neodymium fluoride with the fluorination conversion of about 97% and nearly 100%, respectively. Nd_2O_3 showed higher fluorination conversion than Nd. The mechanism of neodymium oxide transformation to fluoride can be attributed to

(i) chemical reaction with AlF_3 to form NdF_3, as shown in equation (5)
(ii) dissolving in NdF_3 to form $Nd_4O_3F_6$,[16] as shown in equation (6).

$$Nd_2O_3 + 2AlF_3 = Al_2O_3 + 2NdF_3 \quad (5)$$

$$Nd_2O_3\ (s) + 2NdF_3\ (\text{in melt}) = Nd_4O_3F_6\ (\text{in melt}) \quad (6)$$

Obviously, besides chemical reaction, the fluorination of Nd_2O_3 also can achieve by dissolving into the fluoride melt to form oxyfluoride ion, which is beneficial to the fluorination process.

Although NdFeB magnet is dominated by major $Nd_2Fe_{14}B$ phase, also contains Nd rich and $NdFe_4B_4$ phases.[15] The effect of these phases, with different neodymium content and melting temperature, on fluorination process still needs to be clarified by further investigation. As mentioned above, the chemical reaction between Fe, impurities and the fluoride melts to generate iron fluoride was not expected thermodynamically. The differences in chemical activity between Fe, impurities and REEs should be mainly responsible for the selective fluorination process.

Conclusions

In conclusion, REEs in NdFeB magnet scrap have been successfully recovered using AlF_3–NaF melts as the fluorination reagent. The fluorination temperature shows an obvious effect on the rare earth recovery rate. Under the appropriate fluorination conditions, a considerably high recovery rate is expected. The selective fluorination is mainly due to the differences in chemical activity between Fe, impurities and rare earth elements. The obtained fluoride melts can be easily separated from the Fe-based solid residue, and processed with molten salt electrolysis for the direct production of Al–Nd based alloys.

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